

Anatomy and Physiology C - anatomy and physiology! - Monta Vista Science Olympiad (Div. C) - 11-14-2025
Student Answer Sheet (/monta-vista/SM/AnswerSheet?tid=000GI2)

Instructions (shown before students start the test)

Introduction (shown after students start the test)

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(lock in twin you will do great)

1. (1.00 pts) A lesion of the ventral horn at C7 would most predictably cause:

- ☐ A) Ipsilateral loss of vibration sense in the hand
- ☐ B) Ipsilateral flaccid paralysis of wrist/finger extensors
- ☐ C) Contralateral loss of pain and temperature in arm and trunk
- ☐ D) Spastic paralysis of the lower limb

2. (1.00 pts) The Nernst potential for K^+ becomes more negative if:

- ☐ A) $[K^+]_{out}$ increases
- ☐ B) $[K^+]_{in}$ decreases
- ☐ C) $[K^+]_{out}$ decreases
- ☐ D) Temperature increases

3. (1.00 pts) A toxin that prolongs the inactivation of voltage-gated Na^+ channels will most directly:

- ☐ A) Lower AP threshold
- ☐ B) Increase the absolute refractory period
- ☐ C) Increase conduction velocity
- ☐ D) Increase neurotransmitter release probability

4. (1.00 pts) In the adult brain, dopamine from the substantia nigra pars compacta mainly modulates which pathway to facilitate movement?

- ☐ A) Indirect pathway via D2 inhibition
- ☐ B) Direct pathway via D1 activation
- ☐ C) Corticospinal tract via D3 activation
- ☐ D) Reticulospinal pathway via D4 activation

5. (1.00 pts) A lesion of the right optic tract causes:

- ☐ A) Right monocular vision loss
- ☐ B) Left homonymous hemianopia
- ☐ C) Right homonymous hemianopia
- ☐ D) Bitemporal hemianopia

6. (1.00 pts) Rapid adaptation is a hallmark of which receptor?

- ☐ A) Merkel cell (SA1)
- ☐ B) Ruffini ending (SA2)
- ☐ C) Pacinian corpuscle (RA2)
- ☐ D) Muscle spindle (Ia)

7. (1.00 pts) Schwann cells differ from oligodendrocytes because they:

- ☐ A) Myelinate multiple axons each
- ☐ B) Are derived from neural tube
- ☐ C) Participate in axonal regeneration in the PNS
- ☐ D) Form the blood-nerve barrier via tight junctions

8. (1.00 pts) A patient with diabetic autonomic neuropathy is most likely to show which abnormality during Valsalva?

- ☐ A) Exaggerated tachycardia during strain
- ☐ B) Failure of heart rate to increase during phase II
- ☐ C) Excess bradycardia during phase IV
- ☐ D) Hypertensive overshoot during phase II

9. (1.00 pts) CSF is produced primarily by:

- ☐ A) Arachnoid granulations
- ☐ B) Choroid plexus epithelium (modified ependyma)
- ☐ C) Pia mater capillaries
- ☐ D) Astrocytic end-feet

10. (1.00 pts) Bilateral destruction of the amygdala most characteristically causes:

- ☐ A) Profound anterograde amnesia
- ☐ B) Hyperorality and hypersexuality
- ☐ C) Akinetic mutism
- ☐ D) Loss of conditioned motor learning

11. (1.00 pts) The principal excitatory neurotransmitter in the retina from photoreceptors to ON-bipolar cells is:

- ☐ A) Glutamate (sign-inverting via mGluR6 on ON bipolars)
- ☐ B) GABA
- ☐ C) Glycine
- ☐ D) Acetylcholine

12. (1.00 pts) Aqueous humor outflow resistance increases mainly at the:

- ☐ A) Uveoscleral pathway
- ☐ B) Trabecular meshwork
- ☐ C) Ciliary body epithelium
- ☐ D) Canal of Schlemm endothelium

13. (1.00 pts) Taste transduction for "sweet" relies primarily on:

- ☐ A) ENaC channels
- ☐ B) T1R2/T1R3 GPCR → gustducin
- ☐ C) H⁺-gated channels
- ☐ D) Direct Ca²⁺ entry via TRPM5

14. (1.00 pts) Endolymph composition resembles:

- ☐ A) Plasma
- ☐ B) Intracellular fluid (high K⁺, low Na⁺)
- ☐ C) CSF
- ☐ D) Interstitial fluid of cerebellum

15. (1.00 pts) High-frequency sound is detected at the:

- ☐ A) Helicotrema; wide, floppy basilar membrane
- ☐ B) Base; narrow, stiff basilar membrane
- ☐ C) Apex; narrow, stiff basilar membrane
- ☐ D) Oval window; tectorial membrane only

16. (1.00 pts) TRH from the hypothalamus acts via:

- ☐ A) Cytosolic steroid receptor → gene transcription
- ☐ B) RTK → MAPK
- ☐ C) Gq/PLC → IP₃/DAG
- ☐ D) cGMP → PKG

17. (1.00 pts) PTH increases serum Ca^{2+} by all of the following EXCEPT:

- ☐ A) Stimulating osteoclast activation via osteoblast RANKL
- ☐ B) Increasing renal Ca^{2+} reabsorption
- ☐ C) Increasing 1- α -hydroxylase activity
- ☐ D) Decreasing intestinal Ca^{2+} absorption

18. (1.00 pts) Primary hyperaldosteronism typically causes:

- ☐ A) Hyperkalemia, metabolic acidosis, hypertension
- ☐ B) Hypokalemia, metabolic alkalosis, hypertension
- ☐ C) Hyponatremia, hypotension, hyperkalemia
- ☐ D) Hyponatremia, metabolic acidosis, hypotension

19. (1.00 pts) Cortisol exerts permissive effects by:

- ☐ A) Inhibiting 11 β -HSD2 in the kidney
- ☐ B) Upregulating adrenergic receptors on vascular smooth muscle
- ☐ C) Blocking CRH receptors
- ☐ D) Directly opening L-type Ca^{2+} channels

20. (1.00 pts) A pituitary stalk transection acutely causes the least decrease in:

- ☐ A) ACTH
- ☐ B) TSH
- ☐ C) Prolactin
- ☐ D) LH

21. (1.00 pts) Addison's disease is most often associated with:

- ☐ A) Low ACTH, low cortisol, hyperpigmentation
- ☐ B) High ACTH, low cortisol, hyperkalemia
- ☐ C) High ACTH, high cortisol, metabolic alkalosis
- ☐ D) Low ACTH, high cortisol, hypokalemia

22. (1.00 pts) Central diabetes insipidus is best differentiated from nephrogenic DI by:

- ☐ A) Serum sodium
- ☐ B) Response to desmopressin
- ☐ C) Urine specific gravity
- ☐ D) Presence of polyuria

23. (1.00 pts) Graves disease typically shows:

- ☐ A) Low TSI, high TPO Ab, low RAI uptake
- ☐ B) High TSI, high RAI uptake, pretibial myxedema
- ☐ C) Low TSI, low RAI uptake, ophthalmopathy uncommon
- ☐ D) Normal RAI uptake with multinodular pattern only

24. (1.00 pts) A craniopharyngioma most classically causes which visual defect?

- ☐ A) Bitemporal hemianopia
- ☐ B) Right homonymous hemianopia
- ☐ C) Left superior quadrantanopia
- ☐ D) Monocular scotoma

25. (1.00 pts) Broca aphasia is characterized by:

- ☐ A) Fluent speech, poor comprehension
- ☐ B) Nonfluent speech, good comprehension, poor repetition
- ☐ C) Nonfluent speech, poor comprehension
- ☐ D) Fluent speech, good comprehension, poor repetition

26. (1.00 pts) Anterior spinal artery infarct (below T8) spares:

- ☐ A) Pain and temperature
- ☐ B) Motor function
- ☐ C) Dorsal column modalities
- ☐ D) Autonomic sympathetic outflow

27. (1.00 pts) Myasthenia gravis pathophysiology:

- ☐ A) Antibodies to presynaptic Ca^{2+} channels
- ☐ B) Antibodies to postsynaptic nicotinic ACh receptors
- ☐ C) Autoimmune destruction of Schwann cells
- ☐ D) Loss of SNAP-25 function at NMJ

28. (1.00 pts) Lambert-Eaton myasthenic syndrome improves with repeated use because:

- ☐ A) Increased AChR density
- ☐ B) Accumulated presynaptic Ca^{2+} facilitates ACh release
- ☐ C) Closing of K^{+} channels prolongs depolarization
- ☐ D) Increased MuSK signaling

29. (1.00 pts) Brown-Sequard (right hemisection at T10) yields:

- ☐ A) Left loss of vibration below lesion
- ☐ B) Right loss of pain/temp 2 levels below
- ☐ C) Left spastic paralysis below lesion
- ☐ D) Right loss of vibration and motor below lesion

30. (1.00 pts) A glucagonoma would most likely present with:

- ☐ A) Necrolytic migratory erythema, diabetes, weight loss
- ☐ B) Watery diarrhea, hypokalemia, achlorhydria
- ☐ C) Peptic ulcers, diarrhea, steatorrhea
- ☐ D) Walter White

31. (1.00 pts) Cerebellar hemispheric lesions cause:

- ☐ A) Truncal ataxia
- ☐ B) Intention tremor and dysmetria ipsilaterally
- ☐ C) Resting tremor contralaterally
- ☐ D) Hyperreflexia bilaterally

32. (1.00 pts) α -bungarotoxin exposure would most closely mimic:

- ☐ A) Myasthenia gravis
- ☐ B) Lambert-Eaton
- ☐ C) Botulism
- ☐ D) Organophosphate poisoning

33. (1.00 pts) Organ of Corti hair cell depolarization occurs when stereocilia deflect:

- ☐ A) Toward the shortest cilium $\rightarrow$ K^+ efflux
- ☐ B) Toward tallest cilium $\rightarrow$ K^+ influx from endolymph
- ☐ C) Toward tallest cilium $\rightarrow$ Na^+ influx from perilymph
- ☐ D) Toward shortest cilium $\rightarrow$ Ca^{2+} influx from endolymph

34. (1.00 pts) Melatonin is synthesized from:

- ☐ A) Tyrosine
- ☐ B) Tryptophan
- ☐ C) Histidine
- ☐ D) Choline

35. (1.00 pts) SIADH labs classically show:

- ☐ A) Hypernatremia, high serum osmolality, dilute urine
- ☐ B) Hyponatremia, low serum osmolality, concentrated urine
- ☐ C) Hyponatremia, high serum osmolality, dilute urine
- ☐ D) Hypernatremia, low serum osmolality, concentrated urine

36. (1.00 pts) Cushing disease (pituitary) vs ectopic ACTH: which favors ectopic?

- ☐ A) Suppression with high-dose dexamethasone
- ☐ B) Hyperpigmentation absent
- ☐ C) Very high ACTH with refractory cortisol
- ☐ D) Normal DHEA-S

37. (1.00 pts) Hypothyroidism effect on lipids:

- ☐ A) ↓LDL due to increased receptor expression
- ☐ B) ↑LDL and ↑total cholesterol
- ☐ C) ↑HDL only
- ☐ D) ↓Triglycerides only

38. (1.00 pts) Charcot-Marie-Tooth disease most often involves:

- ☐ A) Central demyelination
- ☐ B) PMP22 duplication causing peripheral demyelination
- ☐ C) Autoimmune attack on nodes of Ranvier
- ☐ D) Mutations in dystrophin

39. (1.00 pts) A lesion of the right MLF in internuclear ophthalmoplegia leads to:

- ☐ A) Impaired right eye abduction on right gaze
- ☐ B) Impaired right eye adduction on left gaze with nystagmus of left eye
- ☐ C) Impaired left eye adduction on right gaze with nystagmus of right eye
- ☐ D) Bilateral abduction deficits

40. (1.00 pts) Wernicke encephalopathy triad is due to deficiency of:

- ☐ A) Niacin
- ☐ B) Thiamine
- ☐ C) Pyridoxine
- ☐ D) Cobalamin

41. (1.00 pts) T3 acts at target tissues primarily by:

- ☐ A) Increasing mitochondrial DNA replication directly
- ☐ B) Nuclear receptor binding to TREs to regulate transcription
- ☐ C) Activating adenylate cyclase
- ☐ D) Phosphorylating cytoskeletal proteins directly

42. (1.00 pts) Pheochromocytoma arises from:

- ☐ A) Adrenal cortex zona reticularis
- ☐ B) Neural crest chromaffin cells
- ☐ C) Mesodermal capsule cells
- ☐ D) Pituitary corticotrophs

43. (1.00 pts) Hyperosmolar hyperglycemic state (HHS) vs DKA: which is more typical of HHS?

- ☐ A) Severe acidosis and Kussmaul breathing
- ☐ B) Normal osmolality with ketonemia
- ☐ C) Very high glucose, hyperosmolality, minimal ketosis
- ☐ D) Occurs mostly in type 1 diabetes

44. (1.00 pts) Olfactory pathway uniqueness:

- ☐ A) Projects first to thalamus
- ☐ B) Regenerating receptor neurons from basal cells
- ☐ C) Uses primarily glycine
- ☐ D) Travels exclusively in CN V

45. (1.00 pts) Lesion of Meyers loop causes:

- ☐ A) Right superior quadrantanopia
- ☐ B) Left superior quadrantanopia
- ☐ C) Right inferior quadrantanopia
- ☐ D) Left inferior quadrantanopia

46. (1.00 pts) A neuron has $P_K:P_{Na}:P_{Cl} = 1:0.05:0.45$, with typical ionic gradients. Compared to pure K^+ Nernst, the GHK membrane potential will be:

- ☐ A) More negative (closer to E_K)
- ☐ B) Less negative (depolarized from E_K)
- ☐ C) Equal to E_K
- ☐ D) Unchanged by Cl^- permeability

47. (1.00 pts) A lesion of the lateral hypothalamus would most likely cause:

- ☐ A) Hyperphagia and weight gain
- ☐ B) Aphagia and weight loss
- ☐ C) Hyperthermia
- ☐ D) Diabetes insipidus

48. (1.00 pts) Congenital adrenal hyperplasia due to 21-hydroxylase deficiency presents with:

- ☐ A) Low 17-hydroxyprogesterone, hypertension
- ☐ B) High 17-hydroxyprogesterone, virilization
- ☐ C) High aldosterone, hyponatremia
- ☐ D) Low androgens, ambiguous genitalia in XY

49. (1.00 pts) During dark adaptation, photoreceptors exhibit:

- ☐ A) Low cGMP; $\text{Na}^+/\text{Ca}^{2+}$ channels closed; hyperpolarized
- ☐ B) High cGMP; $\text{Na}^+/\text{Ca}^{2+}$ channels open; depolarized
- ☐ C) High cAMP; K^+ channels open; hyperpolarized
- ☐ D) Low Ca^{2+} ; glutamate release decreased

50. (1.00 pts) A patient has hyponatremia, low cortisol, high ACTH, normal aldosterone and potassium. The most likely diagnosis is:

- ☐ A) Primary adrenal insufficiency
- ☐ B) Secondary adrenal insufficiency
- ☐ C) Isolated ACTH deficiency
- ☐ D) Tertiary adrenal insufficiency from chronic exogenous steroids

51. (3.00 pts) Which statements about homeostatic synaptic scaling are supported by experimental evidence?

(Mark **ALL** correct answers)

- ☐ A) Activity blockade (e.g., TTX) typically induces a divisive down-scaling of AMPAR currents across all synapses on a neuron
- ☐ B) Activity blockade can induce a largely multiplicative up-scaling of AMPAR-mediated miniature EPSCs, preserving relative synaptic weights
- ☐ C) Glia-derived TNF- α is one mediator that can drive scaling of excitatory synapses under reduced activity
- ☐ D) Homeostatic scaling is purely local to individual spines and cannot occur as a global cell-wide process
- ☐ E) Homeostatic scaling serves to maintain a firing-rate set point at the circuit level while allowing Hebbian changes to encode information

52. (3.00 pts) Which statements about intrinsically photosensitive retinal ganglion cells (ipRGCs) are accurate?

(Mark **ALL** correct answers)

- ☐ A) ipRGCs express melanopsin, display peak spectral sensitivity near ~480 nm, and exhibit sustained photoresponses
- ☐ B) ipRGCs can depolarize via a G-protein pathway distinct from rod/cone cGMP-gated currents and contribute to circadian entrainment and the pupil light reflex
- ☐ C) ipRGCs project first to the lateral geniculate nucleus and only secondarily to non-image-forming centers
- ☐ D) Some ipRGC subtypes convey irradiance signals that influence mood and sleep via widespread brain projections
- ☐ E) ipRGC phototransduction has been definitively shown to rely on the canonical rod transduction cascade (transducin → PDE6)

53. (3.00 pts) Which findings regarding the MOC efferent pathway are supported by human/animal data?

(Mark **ALL** correct answers)

- ☐ A) MOC activation reduces cochlear amplifier gain (via outer hair cells), measurable as suppression of otoacoustic emissions
- ☐ B) MOC activity can improve detection in background noise by anti-masking and dynamic range effects
- ☐ C) Acoustic overexposure experiments support a protective role of the MOC reflex against temporary/permanent threshold shifts
- ☐ D) The MOC system exclusively provides ipsilateral efferents; there is no contralateral component in mammals
- ☐ E) Chronic manipulations of afferent drive can potentiate cochlear sensitivity via long-term MOC-dependent plasticity

54. (3.00 pts) Which statements about cortisol pulsatility and HPA-axis dynamics are best supported?

(Mark **ALL** correct answers)

- ☐ A) Normal glucocorticoid signaling is ultradian and circadian, with cell-level decoding of pulse frequency/amplitude critical for gene regulation
- ☐ B) Administration of long-acting glucocorticoids that flatten pulsatility can disrupt physiologic regulation in glucocorticoid-responsive tissues
- ☐ C) Pulsatility is fully explained by SCN-driven circadian input; delayed feedbacks are not required
- ☐ D) Time-delay and feedforward/feedback motifs within the HPA axis can generate ultradian oscillations even without external periodic forcing
- ☐ E) In health, ACTH pulses are typically absent during early sleep

55. (5.00 pts)

FRQ1. Synaptic Integration and Plasticity

Explain how temporal vs spatial summation determines action-potential initiation at the axon initial segment. Include dendritic cable properties, inhibitory shunting (GABA_A), and NMDA receptor coincidence detection in Hebbian LTP.

56. (5.00 pts)

FRQ2. Autonomic Control of the Eye

Contrast the sympathetic and parasympathetic control of pupil and accommodation. Include neurotransmitters/receptors and predict pharmacologic pupil responses to phenylephrine vs pilocarpine in Adie pupil vs CN III palsy.

57. (5.00 pts)

FRQ3. Endocrine Axes and Feedback

Detail HPA axis regulation of cortisol (circadian rhythm, stress, feedback). Compare primary adrenal insufficiency, pituitary Cushing disease, ectopic ACTH, and exogenous steroid use; include renin/aldosterone/K⁺ patterns.

yay you surviveddddd good luck on your other tests!

peace